MTH100

End Semester Exam

Dec 10th, 2024

Time: 2 hours Max Marks: 80

Instructions:

1. Attempt all the eight questions. All questions carry equal marks(parts may have different marks).
2. All your intermediate steps and calculations must be clearly shown.
3. Marks for proof-type questions will depend on the logical progression of the steps. You may quote without proof any proposition or theorem covered in the lectures and tutorials but it must be clearly identified. Any other results used must be proved.

Problem 1. Given the matrix A below.

(a) Find a basis for Nul A and determine the nullity of A.

(b) Find a basis for Col A consisting of columns of A and determine rank(A).

(c) Find a basis for Row A.

$$A = \begin{bmatrix} 2 & 3 & 1 & 1 \\ 1 & -1 & 3 & -1 \\ 1 & 0 & 2 & -1 \\ 3 & 5 & 1 & 2 \end{bmatrix}$$

Problem 2. Find a singular value decomposition (SVD) for the matrix A given below.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 3. Let T be a linear operator on $\mathbb{R}^3$ defined by

$$T(x, y, z) = (-x + 2y + z, y + 3z, x - y + z)$$

(a) Find the matrix of T with respect to the standard ~~matrix~~ basis for $\mathbb{R}^3$, $S = \{e_1, e_2, e_3\}$.

(b) Find the change of basis matrix $P_{S\to\mathcal{B}}$ where $\mathcal{B}$ is the ordered basis.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(c) Find the matrix of T with respect to the the ordered basis $\mathcal{B}$.

(d) Find $[Tv]_{\mathcal{B}}$ where $v = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Problem 4. Let $C^1[0,1]$ be the set of all real valued continuous functions with continuous first derivative.

Define $\langle f, g\rangle = \int_0^1 f'(x)g'(x)dx + f(0)g(0)$

Show that the above satisfies the properties of an inner product on $C^1[0,1]$.

Problem 5. Find a diagonal matrix D and an orthogonal matrix P such that $A = PDP^{-1}$ where

$$A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$ $\frac{1}{5}$

Problem 6. (a) Let A be an $n \times n$ matrix such that $A^2 = A$. Show that the column space of $A = \{\overline{x} \in \mathbb{R}^n : \overline{x} = A\overline{x}\}$.

(b) Show that if P is an orthogonal $m \times m$ matrix, and A is an $m \times n$ matrix, then PA has the same singular values as A.

(c) Let A be an orthogonal 3×3 matrix. Prove that $||A\overline{x}|| = ||\overline{x}||$ for all $\overline{x} \in \mathbb{R}^3$. What are the possible real eigenvalues of $\cancel{A^2}$. A

Problem 7. Let W be the vector space of all real symmetric 2×2 matrices and let $\mathbb{R}_2[t]$ be the vector space of all polynomials with real coefficients of degree ≤ 2.

(a) Find the dimension of W.

(b) Define a linear transformation

$T : W \to \mathbb{R}_2[t]$ by

$$T(\begin{bmatrix} a & b \\ b & c \end{bmatrix}) = (a-b) + (b-c)t + (c-a)t^2$$

Find the rank and nullity of T.

Problem 8. Let $T : V \to W$ be a linear transformation between two finite dimensional vector spaces.

(a) Prove that if dim $V <$ dim W then T can not be onto.

(b) Prove that if dim $V >$ dim W, then T cannot be one-to-one.

(Hint: Use Rank-Nullity Theorem.)